

Mechanical Properties of Solids



Introduction to mechanical properties of solid.

- ∴ Solids have definite shape and volume due to strong intermolecular forces.
- ∴ When an external force is applied solids get deformed (change in shape/size)
- ∴ On removing the force
 - ▶ They regain original shape → Elastic behaviour
 - ▶ They remain deformed → Plastic behaviour
- ∴ The study of how solids respond to external force is called mechanical properties of solids.

* Deforming Force:

A force which changes the size or shape of a body is called a deforming force.

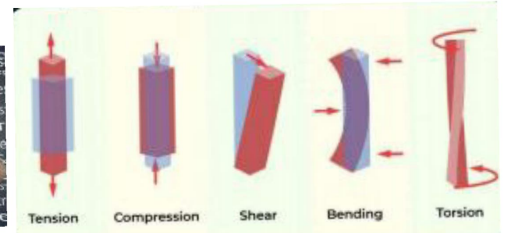
* Elasticity

If a body does not regain its original shape and size even after the removal of deforming force, it is said to be an elastic body and this property is called Elasticity.
Example: Rubber Band, spring

* Plasticity

If a body does not regain its original shape and size even after the removal of deforming force, then it is said to be a plastic body and this property is called Plasticity. Ex-clay, chewing-gum

Stress



Stress:
The internal restoring force set up per unit area of cross-section of the deformed body is called

Stress.

$$\sigma = \frac{\text{Force}}{\text{Area}} \quad \sigma = \frac{F}{A} \text{ - Nm}^{-2}$$

$$\text{SI unit} = \text{Nm}^{-2}$$

Dimensional [ML⁻¹T⁻²]
Formula

$$\frac{\text{N}}{\text{m}^2} = \frac{\text{kgm}}{\text{s}^2 \text{m}^2} = \frac{\text{kg}}{\text{ms}^2} \quad \text{Pressure} = \frac{\text{Force}}{\text{Area}} \quad \text{Pascal}$$

$$= \text{kgm}^{-1}\text{s}^{-2}$$

Types of stress

1 Tensile stress

It is the restoring force set up per unit cross-sectional area of a body when the length of the body increases in the direction of the deforming force. It is also known as longitudinal stress.

2 compressional stress:

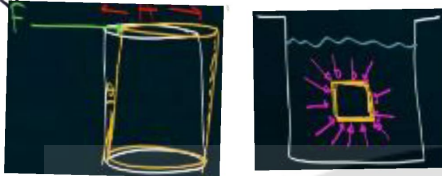
It is the restoring force set up per unit cross-sectional area of a body when its length decreases under a body when its length decreases under a deforming force.

3 Hydrostatic stress

If a body is subjected to a uniform force from all sides, then the corresponding stress is called Hydrostatic Stress.

4 Tangential or Shearing Stress

When a deforming force acts tangentially to the surface of the surface of the body, it produces a change in the shape of the body. The tangential force applied per unit area is equal to the tangential stress.



Strain

The ratio of change in any dimension produced in the body to the original dimension is called strain.

$$\text{Strain } (\epsilon) = \frac{\text{change in dimension}}{\text{original dimension}}$$

★ unitless

★ Dimension

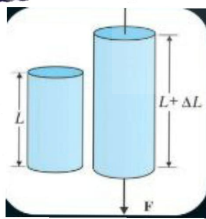


Types of strain

1 Longitudinal strain

It is defined as the increase in the length per unit original length, when the body is deformed by external forces.

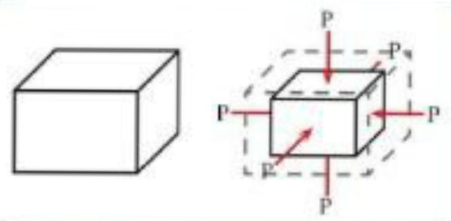
$$\text{Longitudinal strain} = \frac{\text{change in length}}{\text{original length}} = \frac{\Delta L}{L}$$



2 Volumetric strain

It is defined as the change in volume per unit original volume when the body is deformed by external force.

$$\text{volumetric strain} = \frac{\text{change in length}}{\text{original length}} = \frac{\Delta V}{V}$$



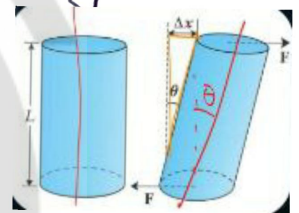
3 Shear strain:

It is defined as the angle (in radian) through which a face originally perpendicular to the fixed face gets turned on applying tangential deforming force.

$$\text{Shear strain} = \theta = \tan \theta = \frac{\text{Relative Displacement B/w 2 parallel planes}}{\text{Between 2 parallel planes}}$$

$$\tan \theta = \frac{\Delta x}{L} \quad \text{Q} \rightarrow \text{Bohot chota angle for small angle } \tan \theta \approx \theta$$

$$\theta = \frac{\Delta x}{L}$$



Hooker's Law

For small deformations the stress and strain are proportional to each other. This is known as Hooke's law.

Stress & strain

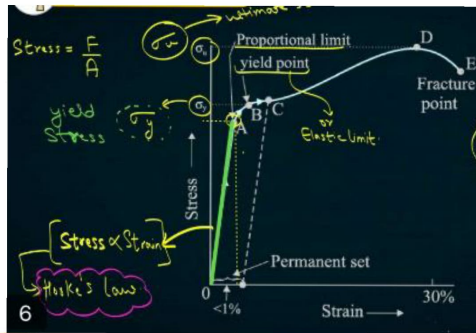
$$\text{or Stress} = \text{constant} \times \text{strain}$$

$$\text{or } \frac{\text{Stress}}{\text{Strain}} = \text{constant}$$

Where K is the proportionality constant and is known as modulus of elasticity.



Stress - Strain curve



Note

Hooke's law is only valid till Proportional limit

1 Proportional Limit

- ∴ Stress is directly proportional to Strain (Hooke's law holds)
- ∴ curve is linear

2 yield point or Elastic Limit (EL):

- ∴ maximum stress up to which the metal returns to its original shape after removing the load
- ∴ yield strength = stress at which permanent deformation begins.

3 plastic Region:

- ∴ Beyond yield point, permanent deformation occurs.
- ∴ If load is removed here, metal exhibits permanent set (retains new shape)

4 ultimate stress or ultimate Tensile Strength:

- ∴ maximum stress the metal can withstand.
- ∴ Beyond this, necking starts (cross-sectional area reduces)

5 Fracture point (F):

- ∴ metal breaks into two pieces

Permanent set:

- ∴ The residual deformation that remains after removing the

load in the in the plastic region

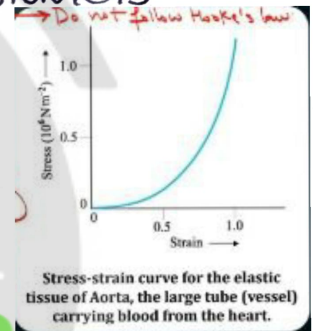
Ductile material:

- ∴ Can undergo large plastic deformation before fracture (eg mild steel).
- ∴ shows a long plastic region

Brittle material

- ∴ fractures without significant plastic deformation (eg cast iron)
- ∴ very short or almost no plastic region.

Substances like tissue of aorta, rubber etc, which can be stretched to cause large strains are called elastomers



Elastic Moduli

- ∴ Different types of moduli of elasticity. corresponding to the three types of strain we have three important moduli of elasticity

- Young's modulus (γ) i.e. the modulus of elasticity of length
- Bulk modulus i.e. the modulus of elasticity of volume
- Modulus of rigidity or shear modulus (η) i.e. modulus of elasticity of shape

$$\frac{\text{Stress}}{\text{Strain}} = \text{Elastic Modulus}$$

unit - N/m^2

$[\text{ML}^{-1}\text{T}^{-2}]$

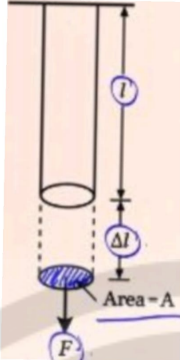
Young's modulus

∴ The ratio of tensile (or compressive) stress (σ) to the longitudinal strain (ϵ) is defined as young's modulus and is denoted by the symbol γ

$$\gamma = \frac{\text{Longitudinal stress } \text{N/m}^2}{\text{Longitudinal strain unitless}}$$

$$\gamma = \frac{F/A}{\Delta l/l} = \gamma = \frac{Fl}{AA\Delta l}$$

unit - N/m^2 or pascal
dimension - $[ML^{-1}T^{-2}]$

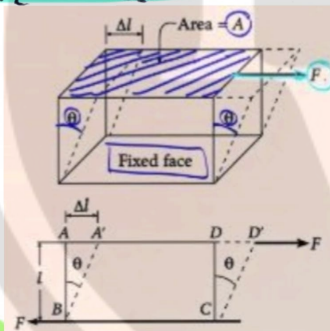


Shear modulus

∴ modulus of rigidity or shear modulus within the elastic limit, the ratio of tangential stress to shear strain is called modulus of rigidity

$$G = \frac{\text{Tangential stress } = F/A}{\text{Shear strain } = \theta} \text{ radian}$$

$$G = \frac{F}{A\theta}$$



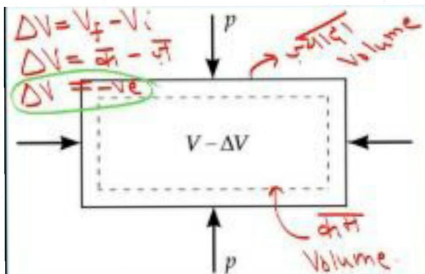
Bulk modulus

∴ The ratio of hydraulic stress to the corresponding hydraulic strain is called bulk modulus. It is denoted by symbol B

$$B = \frac{\text{Hydraulic stress } = \frac{F}{A}}{\text{Hydraulic strain } = -\frac{\Delta V}{V}}$$

$$B = \frac{p}{-\Delta V/V} \Rightarrow B = -\frac{pV}{\Delta V}$$

unit pascal (pa), Where p is the pressure



Compressibility

∴ The reciprocal of the bulk modulus is called compressibility and is denoted by K.

$$K = \frac{1}{B} \Rightarrow K = \frac{1}{-\frac{pV}{\Delta V}} \Rightarrow \boxed{K = -\frac{\Delta V}{pV}}$$

Note Shear modulus (G) is always less than young's modulus (γ)
Generally $G = \frac{\gamma}{3}$

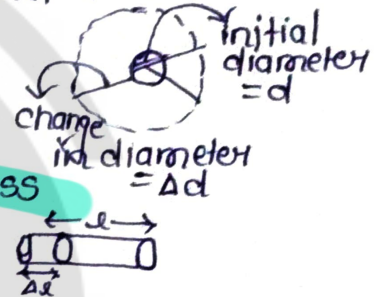
Poisson's Ratio

Poisson's Ratio is defined as the negative of ratio of lateral strain to the longitudinal strain

$$\nu = -\frac{\Delta d/d}{\Delta l/l}$$

→ unitless

→ dimensionless



Elastic potential Energy in a stretched wire



$$W = \int F \cdot dl$$

$$\Rightarrow W = \int \frac{\gamma A l}{L} \cdot dl$$

$$\Rightarrow W = \frac{\gamma A}{L} \int l \cdot dl$$

$$\Rightarrow W = \frac{\gamma A}{L} \left[\frac{l^2}{2} \right]$$

$$U = \frac{1}{2} \frac{\gamma A}{L} l^2$$

→ Elastic PE

$$\Rightarrow U = \frac{1}{2} \times \underbrace{\sigma}_{\text{Stress}} \times \underbrace{\epsilon}_{\text{Strain}} \times \underbrace{V}_{\text{Volume}}$$

→ This work done is stored in the form of Elastic PE

$$\gamma = \frac{F/A}{\Delta l/l} = \frac{FL}{A\Delta l}$$

$$\Rightarrow \left[F = \frac{\gamma A \Delta l}{L} \right]$$

Elastic potential density

$$\Rightarrow U = \frac{W}{V}$$

$$\Rightarrow U = \frac{1}{2} \sigma \epsilon$$